Carryover representation and inference for biomarker-treatment interaction tests in aggregated N-of-1 trials under covariance moderation: a staged simulation study and a recommended analysis for the Hybrid design

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Contents

Abstract **2**

1 Introduction **3**

2 Simulation design **5**

2.1 Aims 6

2.2 Trial designs 6

2.3 Data-generating mechanism 7

2.4 Estimand and analysis specifications 7

2.5 Choice of analysis specifications 11

2.6 Factorial grid and parameter settings 11

2.7 Performance measures and computing environment 12

2.8 Corrections to the simulation code during the study 12

3 Results **13**

3.1 Stage 1: representing residual exposure 13

3.2 Stage 2: removing the analyst's half-life 15

3.3 Stage 3: a correctly sized test 16

3.4 Stage 4: external comparators on the same trials 17

3.5 Stage 5: prognostic confounds, feasibility, and random effects 20

4 Discussion **23**
4.1 What each stage established 23
4.2 The recommended analysis for a Hybrid trial 24
4.3 What the expectancy terms are for 25
4.4 Limitations 25
4.5 Related approaches not evaluated 25

5 Conclusions **26**

6 Declarations **26**

7 Reproducibility **27**

8 References **28**

Abstract

Background. Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Under covariance moderation, a multivariate-normal differential-correlation architecture in which the biomarker-treatment interaction lives in a treatment-state-dependent correlation, a companion study found that carryover costs up to roughly 31% of power in relative terms. The analyst must still decide how to represent residual exposure in the linear predictor, how to obtain a correctly sized test, and whether to protect the test against a biomarker that is prognostic rather than predictive.

Methods. We build the analysis model in stages, each stage answering one question the previous one raised, within an ADEMP-structured simulation[1] of the refined-process data-generating mechanism established in the companion study. Stage 1 asks how residual drug exposure should enter the linear predictor, comparing the binary on-drug indicator (G1), that indicator plus a classical lagged-treatment term (G2) and a continuous exposure-weighted predictor that decays with time since discontinuation at an assumed half-life (G3), across a full factorial of decay form, trial design, sample size and interaction strength (216 cells, 500 replicates each). Stage 2 removes the analyst's assumed half-life by choosing it from the data by AIC (G4). Stage 3 corrects the size of the test: the model-based standard error is conservative under the mis-specified working covariance, so each specification is refitted with a CR2 bias-reduced cluster-robust standard error and Satterthwaite degrees of freedom (G8, G9). Stage 4 compares the resulting family against two external comparators on the same simulated trials: the originally published analysis, which has no residual correlation structure (E0), and a richer family that adds design-expectancy terms and either fixes or bootstrap-calibrates the half-life (F1, F1A0B). Stage 5 stress-tests the leading specification against four prognostic biomarker pathways at two confound strengths and against richer random-effects

structures. Two defects in the simulation code found during the study were corrected and every comparison reported here was run under the corrected code.

Results. The lagged-treatment term adds nothing: G2 and G1 differ by a mean of 0.002 in power across all 216 cells and agree on every other measure. Exposure weighting recovers most of the power that carryover removes, by 8 points at the Hybrid reference cell (0.888 against 0.806 for G1 at $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1$ week) and 11 points under OL+BDC, and is never worse than G1 in any design; an apparent collapse under the crossover design in earlier drafts was the coding defect. Choosing the half-life by AIC matches the oracle-half-life predictor without requiring the analyst to know the half-life. The model-based test rejects at 0.037 against nominal 0.05; CR2 restores nominal size for every specification. On the corrected 30-cell grid, G9 (AIC-selected half-life plus CR2) is the most powerful or tied most powerful specification at every Hybrid and OL+BDC alternative cell, with Type I error within the Monte Carlo band at all eight null cells. The published analysis E0 is liberal at every null cell (0.06 to 0.15). G9 holds nominal size under all four prognostic pathways without an expectancy term, at both confound strengths, and is at least as powerful as the bootstrap-calibrated F1A0B in every alternative cell, including a four-fold half-life misspecification. A power loss at the strongest confound, seen by every specification, was traced to the confound being infeasible: the joint covariance matrix was not positive definite and had been silently repaired. A closed-form feasibility ceiling now bounds such scenarios. Random slopes on time or expectancy change G9's power by under one point.

Conclusions. For a Hybrid aggregated N-of-1 trial under covariance moderation we recommend G9: the exposure-weighted predictor with its half-life chosen by AIC on the null model and refitted by REML, tested with a CR2 cluster-robust standard error and Satterthwaite degrees of freedom. It requires no pharmacokinetic input from the analyst, holds nominal size including under prognostic biomarkers and after half-life selection, and loses no power to the richer expectancy-term family. The classical lagged-treatment term should not be reported as a carryover adjustment. Open items are a dropout check for G9 specifically and a rerun of the exploratory sensitivity blocks of earlier drafts under the corrected code. The scope is the covariance-moderation architecture.

1 Introduction

We begin with the basic design. In a single N-of-1 trial, one patient is crossed repeatedly between active treatment and a comparator. The within-patient contrast between on-drug and off-drug periods estimates that patient's individual treatment effect. An aggregated N-of-1 trial pools many such series, which allows for population-level inference at sample sizes smaller than a parallel-group trial would require[2]. The estimand is the biomarker-treatment interaction, that is, whether a baseline covariate B_i modifies the treatment effect. We evaluate analysis approaches for inference regarding this estimand in a 'hybrid' N-of-1 design, defined in Section 2.2.

The main inferential complication in N-of-1 trials is carryover. A drug's pharmacological effect does not switch off abruptly at discontinuation. It continues to decay as a function of the

drug's half-life, so post-discontinuation observations remain somewhat indicative of treatment effect. An exposure regressor that ignores this is, in effect, a mis-measured version of what the patient actually received, and the damage is twofold. First, the interaction contrast is diluted. Occasions still carrying a substantial fraction of the drug effect are coded identically to occasions carrying none, which biases the biomarker-by-exposure coefficient toward zero, as any covariate measured with error would. Second, those same occasions are heterogeneous in a way the model does not represent, so the unexplained variation inflates the standard error of the very coefficient that has just been attenuated. The test loses on both counts, and power falls accordingly.

We note that this is a loss of power, not of validity. Under the null, $c_{bm} = 0$, no biomarker-response association exists at any level of residual exposure, so a mis-coded regressor has no interaction to distort, and test size is unaffected by exposure coding. The conservatism observed in Section 3.3 has a separate origin in the working covariance and applies equally across specifications. Carryover mis-specification is therefore a question of efficiency, and the first stage of the study is framed in terms of recovered power.

A companion study[3], building on Hendrickson[2], shows that under the covariance-moderation architecture carryover erodes power directly and design-dependently. In that architecture the interaction lives in the treatment-state-dependent covariance $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$. At a half-life of one week, the relative loss is roughly 31% under OL+BDC but only 3 to 5% under crossover and Hybrid. We restrict attention here to covariance moderation, where the carryover problem is largest and where Hendrickson's original results were obtained. That earlier work fixed the decay shape at exponential and evaluated only one analysis specification.

This manuscript reports the process by which we arrived at a recommended analysis for the Hybrid design, rather than a single cross-sectional comparison. The analysis model was built in stages, each stage answering a question the previous one raised, and the order of the stages is the order in which the questions arise for an analyst.

The first question is how residual exposure should enter the linear predictor. Three positions are available toward the unknown decay. The analyst may deny it, coding exposure strictly on or off (Unadjusted, G1), which is the benchmark any remedy must beat. The analyst may estimate it, adding a lagged just-off-drug nuisance term whose magnitude the data determine (Lag-adjusted, G2), the classical crossover-trial device of Jones and Kenward[4]. Or the analyst may assume it, committing to a decay form and half-life in a continuous exposure-weighted predictor (Exposure-weighted, G3). G1 and G2 estimate the same coefficient and differ by exactly one nuisance term, so their contrast isolates that term; G3 changes the regressor carrying the interaction. We cross all three against exponential and Weibull data-generating decay forms in a full factorial (Stage 1).

The second question follows from the third position: an assumed half-life is an oracle no real analyst has. We therefore let the data choose it by AIC over a candidate grid (G4) (Stage 2).

The third question concerns the size of the test rather than its power. The working covariance of the mixed model, a random intercept with an AR(1) residual correlation, does not match the covariance the data-generating mechanism induces, and the model-based test is conservative.

A CR2 bias-reduced cluster-robust standard error with Satterthwaite degrees of freedom[5] corrects this, and the correction is applied to each specification, with a dedicated check that data-driven half-life selection does not inflate size (Stage 3).

The fourth question is whether anything outside this family does better. Two comparators are fitted to the same simulated trials: the originally published analysis, which uses a binary indicator and no residual correlation structure (E0), and a richer family developed in parallel within this program that adds design-expectancy terms to protect against a prognostic biomarker and either fixes the half-life or calibrates its selection by a parametric bootstrap (F1, F1A0B) (Stage 4).

The fifth question is robustness of the leading specification to the conditions a real trial is most likely to violate: a biomarker that is prognostic as well as predictive, through four distinct pathways and at two strengths, and a random-effects structure richer than a random intercept (Stage 5). This stage also produced a methodological finding about the simulation itself, a silently repaired non-positive-definite covariance matrix, which we report because it changes how such stress tests must be designed.

Two defects in the simulation code were found during the study and corrected: a pre-exposure coding error in the exposure-weighted predictor that affected the crossover design, and a cross-factor correlation constant that departed from the documented data-generating process. Every result in this manuscript was produced under the corrected code, with one labeled exception. We follow the ADEMP reporting framework of Morris, White, and Crowther[1] throughout. Section 2 describes the simulation design in ADEMP order, Section 3 gives the results stage by stage, and Section 4 states the recommendation and its limits.

2 Simulation design

The study simulates complete aggregated N-of-1 trials and analyzes each one under every specification. A single replicate proceeds as follows. Patients are allocated across the randomization paths of the chosen trial design (Section 2.2). For each patient we draw a pre-treatment biomarker together with the three latent response components, namely the pharmacological response, the natural-history trend, and the placebo-belief component. These are drawn jointly across all eight measurement occasions from a multivariate normal distribution whose means follow modified Gompertz trajectories. The covariance carries within-component AR(1) autocorrelation, cross-component correlations, and the treatment-state-dependent biomarker-response correlation that encodes the interaction (Section 2.3). The observed symptom score at each occasion is the patient's baseline severity less the sum of the three components, so that a beneficial effect lowers the score. Carryover enters through a decay function of time since discontinuation, which keeps part of the pharmacological effect present at nominally off-drug occasions. Every analysis specification in a given stage is then fitted to that same simulated dataset, and we record the interaction coefficient, its standard error, and its p-value from each. This is repeated 500 times for every parameter combination in the grid of Section 2.6, and 1,000 times in the random-effects comparison of Stage 5.

Fitting all specifications to a common dataset is deliberate. It makes every contrast among them paired, so that differences are estimated with substantially less Monte Carlo variance than independent draws would give. Throughout, N denotes the **total** number of patients in a trial, allocated as evenly as possible across the design's randomization paths, so a design with P paths assigns about N/P patients to each. At $N = 70$ this is 35 per path under the two-path designs and 17 or 18 per path under the four-path Hybrid design.

2.1 Aims

The aim is to arrive at a recommended analysis specification for the biomarker-treatment interaction in a Hybrid aggregated N-of-1 trial under covariance moderation, and to document the evidence at each stage of its construction. The stages ask, in order: how residual exposure should enter the linear predictor; whether the analyst's assumed half-life can be replaced by a data-driven choice; whether the test is correctly sized and how to make it so; whether the originally published analysis or a richer expectancy-term family does better on the same trials; and whether the leading specification survives a prognostic biomarker and a richer random-effects structure.

2.2 Trial designs

The three designs differ in how on-drug and off-drug occasions are arranged, and so in how much post-discontinuation information each generates. All three schedule eight measurement occasions per patient across randomization paths, each a fixed on-drug/off-drug schedule. Each design also carries an expectancy weight, one during open-label periods and one half during blinded periods, scaling the placebo-belief component.

Crossover (CO). A traditional two-period within-patient crossover, blinded throughout, occasions spaced 2.5 weeks apart from 2.5 to 20 weeks. One path is active for the first four occasions then comparator, and the other reverses that order and so never discontinues, contributing no post-discontinuation observations.

Open-label followed by blinded discontinuation (OL+BDC). Open-label titration at 4, 8, 12, and 16 weeks, then weekly follow-up to week 20 with a blinded late switch to placebo. The two paths discontinue after week 17 or 18, so only two or three occasions follow.

Hybrid. The Hendrickson design and this program's reference. Open-label titration at weeks 4 and 8 is followed by blinded discontinuation weekly from week 9 to 12, then a brief two-occasion crossover at weeks 16 and 20. The four paths vary discontinuation timing and crossover-arm order. It is the richest of the three in on-off transitions and the only one allowing more than one discontinuation per patient. It is the design for which this manuscript's recommendation is made; CO and OL+BDC are carried through Stages 1 to 4 to show where the recommendation does and does not transfer.

One consequence bears directly on the results. The first post-discontinuation occasion falls one week after switching under OL+BDC and Hybrid, but 2.5 weeks under CO. At the half-lives

examined here, roughly half the pharmacological effect therefore persists into the first off-drug occasion under the two non-crossover designs, against less than a fifth under CO. The designs therefore differ substantially in how much carryover contamination they can accumulate at all.

2.3 Data-generating mechanism

Under covariance moderation, the biomarker and the biological response are drawn jointly from a multivariate normal distribution with a treatment-state-dependent correlation, $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form and t_{sd} the time since drug discontinuation. The data-generating process is the refined process of the companion study: graded coupling (zero before first exposure), separable AR(1) within-component correlation at $\rho = 0.7$ with a single cross-component constant $c_1 = 0.2$ scaled by $\rho^{|w_t - w_s|}$ at every lag, anchored carryover on the drug-response mean, and a stop on non-positive-definite covariance matrices (Section 2.8). Mean moderation appears only as one scenario cell in Stage 4; the full cross-architecture comparison is in the archived full-scope drafts (`archive/report.Rmd`, `archive/report-slim.Rmd`).

The decay function $\phi : \mathbb{R}_+ \to [0, 1]$ maps t_{sd} to a residual-effect multiplier. Two forms are used, each matched to a common half-life $t_{1/2}$. Exponential decay,

$$\phi_{\exp}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}},$$

is classical first-order elimination, the default pharmacokinetic assumption and the form every analysis specification assumes. Weibull decay,

$$\phi_{\mathrm{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}},$$

has a shape parameter k governing whether the elimination tail is heavier than exponential ($k < 1$) or lighter ($k > 1$), recovering the exponential form at $k = 1$. The Stage 1 factorial uses $k \in \{0.7, 1.5\}$ alongside exponential; the Stage 4 decay-shape misspecification cell uses $k = 2$. All forms pass through the same half-life-defining point, $\phi = 0.5$ at $t_{sd} = t_{1/2}$, so they are comparable at a fixed nominal half-life even though their shapes diverge on either side of it.

2.4 Estimand and analysis specifications

The interaction estimand is the on-drug biomarker-response correlation c_{bm}, a dimensionless quantity. Bias and empirical SE are reported on a calibrated effect-size scale, namely the expected difference in on-drug response per SD of the biomarker at $t_{1/2} = 0$. Power and Type I error are on the native test-statistic scale, with the null estimand $c_{bm} = 0$. The base analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation`

Table 1: The nine G-family analysis specifications. G1-G3 are the Stage 1 representations of residual exposure; G4 replaces the assumed half-life by a data-driven choice (Stage 2); G6-G9 are CR2 cluster-robust variants of G1, G2, G4 and G3 (Stage 3). G5 collapses to one row per patient before fitting and has no CR2 variant. The external comparators E0 and the F-family are defined in Section 2.4.4.

Code	Stored	Name	Formula	Target coefficient
G1	E1	Unadjusted	Sx bm + t + Db + bm:Db	bm:Db
G2	E3	Lag-adjusted	Sx bm + t + Db + bm:Db + L	bm:Db
G3	E2	Exposure-weighted	Sx bm + t + Dbc + bm:Dbc	bm:Dbc
G4	E7	AIC-selected t1/2	G3 formula, t1/2 by AIC over {0.25,0.5,1.0,2.0}	bm:Dbc
G5	E9	Paired-difference	diff bm (patient-level OLS)	bm slope
G6	E1cr2	Unadjusted + CR2	G1 formula, CR2 SE	bm:Db
G7	E3cr2	Lag-adjusted + CR2	G2 formula, CR2 SE	bm:Db
G8	E2cr2	Exposure-weighted + CR2	G3 formula, CR2 SE	bm:Dbc
G9	E7cr2	AIC-selected t1/2 + CR2	G4 formula, CR2 SE	bm:Dbc

`= corCAR1(form = ~ t | ptID))`. The specifications differ in how X is constructed, whether a nuisance term is added, and how the standard error is obtained. Internally, and in the code and data files described in the Reproducibility section, each specification is stored under an `E`-prefixed code; this manuscript refers to them by the compact `G1-G9` code of Table 1, or by name. Six of the nine carry the argument: G1, G2 and G3 (Stage 1), G4 (Stage 2), and G8 and G9 (Stages 3 to 5). G5, G6 and G7 appear in the nine-specification heatmaps of Stage 3 for completeness and are otherwise not discussed.

2.4.1 Unadjusted (G1): binary on-drug indicator

Here $X = D_{it}$, the binary on-drug indicator (one on drug, zero off). Carryover is not represented at all. An off-drug occasion still carrying most of the pharmacological effect is therefore coded identically to one carrying none. This is the default an analyst obtains without considering carryover, and we treat it as the benchmark the other specifications must justify their additional structure against. It consumes no half-life, so it is invariant to the analyst's pharmacokinetic assumptions.

2.4.2 Lag-adjusted (G2): binary indicator plus estimated carryover term

We retain $X = D_{it}$ and augment the model with a lagged-treatment indicator, $L_{it} = D_{i,t-1}(1 - D_{it})$, marking off-drug timepoints that immediately follow an on-drug one. This is the classical crossover-trial device of Jones and Kenward[4]. Carryover magnitude is estimated from the data rather than assumed, so, like Unadjusted, this specification is invariant to the assumed half-life. L_{it} enters only as a nuisance main effect. There is no bm:L_{it} product, so the interaction actually tested remains bm:D_{it}. Lag-adjusted can absorb mean-level carryover, but it cannot repair the interaction's attenuation.

2.4.3 Exposure-weighted (G3): continuous decayed predictor

Here $X = D_{bc,it}$, a continuous exposure-decayed predictor matched to the analyst's assumed decay form and half-life. At on-drug timepoints $D_{bc,it} = 1$. Off drug after first exposure it decays as $D_{bc,it} = (1/2)^{t_{sd,it}/\hat{t}_{1/2}}$, losing half its value every $\hat{t}_{1/2}$ weeks after discontinuation. Before any exposure $D_{bc,it} = 0$; this pre-exposure guard was absent in earlier drafts (Section 2.8). The half-life assumption enters only through the predictor values handed to `lme`, so Exposure-weighted is, in effect, a different model for each assumed half-life. As $\hat{t}_{1/2} \to 0$, $D_{bc,it} \to D_{it}$, so Unadjusted is the boundary member of the Exposure-weighted family at zero assumed half-life. In Stage 1 the assumed half-life is set equal to the data-generating one, an oracle no real analyst has; Stage 2 removes it.

2.4.4 AIC-selected half-life (G4)

This specification uses Exposure-weighted's formula but chooses $\hat{t}_{1/2}$ from the data. Writing $M(t_{1/2})$ for the `lme` fit with $D_{bc,it}$ constructed at a candidate half-life, the selected half-life is

$$\hat{t}_{1/2} = \underset{t_{1/2} \in \mathcal{T}}{\arg\min}\, \mathrm{AIC}(M(t_{1/2})), \qquad \mathcal{T} = \{0.25,\ 0.5,\ 1.0,\ 2.0\} \text{ weeks},$$

with $\mathrm{AIC}(M) = -2 \log L(M) + 2p$[6]. The candidate fits are compared by maximum likelihood, since the candidates differ in a fixed-effects column and REML likelihoods are not comparable across fixed-effects designs; the selected model is then refitted by REML and the interaction coefficient, its standard error and p-value are read from that single refit, not pooled across $\mathcal{T}$. This mirrors `characterize_carryover()` (`R/carryover_analysis.R`), the package's own real-data half-life-selection routine. The cost is a post-selection inference risk[7]: the reported standard error treats $\hat{t}_{1/2}$ as fixed in advance when the same data chose it. Stage 3 checks whether this manifests as Type I error inflation.

2.4.5 CR2 cluster-robust variants (G8, G9)

The standard error of $\beta_{bm:X}$ read from the fitted `lme` object is valid only if the assumed residual covariance is correct. The working covariance combines a patient random intercept with a `corCAR1` AR(1) residual correlation, and the companion calibration study[8] shows this does not match the covariance the data-generating mechanism induces. We quantify the consequence with the calibration ratio

$$\kappa = \frac{\overline{\widehat{SE}}}{SD(\hat{\beta}_{bm:X})},$$

the mean model-reported standard error across replicates divided by the empirical standard deviation of the replicate point estimates; $\kappa = 1$ is correctly calibrated and $\kappa > 1$ a conservative test. The remedy is a cluster-robust sandwich variance, which derives the variance

of $\hat{\beta}_{bm:X}$ from observed between-patient variation rather than the assumed covariance. With only $N = 35$ or 70 clusters the naive Liang-Zeger form is itself downward-biased, so we use the CR2 bias-reduced estimator of Bell and McCaffrey[5] with Satterthwaite degrees of freedom, via `clubSandwich`, replacing only the standard error, reference distribution and p-value and leaving point estimates untouched. G8 is G3 with CR2 inference and G9 is G4 with CR2 inference; G6 and G7 are the corresponding variants of G1 and G2.

2.4.6 External comparators: the published analysis (E0) and the F-family

Two further analyses are fitted alongside the G-family to the same simulated trials in Stages 3 to 5.

E0 is the analysis actually published with the data-generating process[2], read from the original repository (commit `58b32a9`, `R/lme_analysis.R`) rather than from this project's vendored copy, which had acquired a residual correlation structure at some later point. It fits `lme4::lmer` with a random intercept and no residual correlation structure, uses the binary on-drug indicator, and takes Satterthwaite p-values from `lmerTest`. Every G-family specification therefore differs from it in carrying `corCAR1`, which is a project addition.

The F-family adds the design-expectancy covariate D_e (η_t, one during open-label periods and one half during blinded periods) and its product with the biomarker to G3's formula, $\text{Sx} \sim bm + t + D_{bc} + bm{:}D_{bc} + D_e + bm{:}D_e$, with the same random intercept, `corCAR1` and CR2 inference as G8 and G9. The expectancy terms were introduced in the companion study to protect the interaction test against a biomarker that is prognostic rather than predictive. The members differ only in how the analysis half-life is set: **F1** fixes it at a policy value of 0.2 week; **F1A** selects it by AIC with the interaction in the selection model; **F1A0** selects it on the null model and refits with the interaction, as G4 does; **F1A0B** calibrates F1A0's test by a parametric bootstrap that repeats the selection on every resample. F1 and F1A0B are the two reported here. Because D_e is constant within every path of the crossover design, the F-family is not identifiable there and is reported for Hybrid and OL+BDC only.

2.4.7 Prognostic biomarker pathways

A biomarker may be associated with the outcome through channels other than the drug response, and an interaction test that is fooled by such an association is not usable in practice. Four pathways are implemented in the data-generating process, each setting one parameter with all others at their defaults: a correlation between the biomarker and baseline severity (prog-BL, c_{bl}); a correlation with the natural-history component at every occasion (prog-TV-const, c_{tv}); an additive shift of the natural-history mean proportional to the standardized biomarker (prog-TV-grow, b_{tv}); and a correlation with the placebo-response component at every occasion (prog-PB, c_{pb}). Three of the four are entries in the biomarker row of the joint correlation matrix, and their feasible strengths are bounded; Stage 5 derives the bound.

2.5 Choice of analysis specifications

The specifications were chosen to represent distinct, established positions an analyst can take toward an unmeasured decay, and then to answer the specific questions each stage raised, not to survey the wider space of N-of-1 analysis models. Bayesian hierarchical models, GEE as a primary model, randomization tests, flexible spline decay curves, state-space treatments of drug concentration and two-stage meta-analytic combination are all established in the N-of-1 or crossover literature and none is evaluated here; Section 4.5 returns to three of them. An earlier draft also tested carryover terms interacted with the biomarker (`bm:L` and `bm:t_{sd}`); both underperformed G1 at every cell, which we attribute to the data-generating process having a single true decay parameter, so that a second interacted coefficient adds estimation noise with no signal to recover. They were dropped in favor of G4, which addresses the same unknown, the decay half-life, by estimating it.

2.6 Factorial grid and parameter settings

Stage 1 crosses biomarker moderation, carryover half-life, trial design, sample size and decay form against G1, G2 and G3:

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Sample size (N)	35, 70
Trial design	CO, Hybrid, OL+BDC
Decay form	exponential; Weibull $k \in \{0.7, 1.5\}$
Replicates per cell	500

Three half-lives, three designs, two sample sizes, three biomarker levels and three decay forms give 162 exponential-plus-Weibull cells in the covariance-moderation slice reported here (216 cells counting the $k = 1.0$ Weibull condition, which duplicates exponential numerically and is retained in the production grid). The $c_{bm} = 0$ row is the Type I error check against nominal 5%, and the $t_{1/2} = 0$ row gives a best-case power reference at each combination of N, design and c_{bm}.

Stages 2 to 4 use a 30-cell grid on which every specification is fit to the same trial each replicate: exponential decay, design $\times$ {CO, Hybrid, OL+BDC}, $N \in \{35, 70\}$, with Panel A varying $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1$ week and Panel B varying $t_{1/2} \in \{0, 0.5\}$ at $c_{bm} = 0.45$. Stage 4 adds fifteen scenario cells on Hybrid at $N = 70$ and $t_{1/2} = 0.5$ week: the null and two alternatives ($c_{bm} = 0.45$, 0.25); OL+BDC null and alternative; the four prognostic pathways at strength 0.3 under the null; a four-fold half-life misspecification (true $t_{1/2} = 2$ weeks, null and $c_{bm} = 0.25$); a decay-shape misspecification (Weibull $k = 2$ data, exponential analysis, null and $c_{bm} = 0.25$); and the mean-moderation architecture (null and $c_{bm} = 0.25$). Stage 5 crosses the four pathways with strength $\{0.3, 0.6\}$, $c_{bm} \in \{0, 0.45\}$ and

$N \in \{35, 70\}$ on Hybrid at $t_{1/2} = 0.5$ week (32 cells, 500 replicates), and refits G9 with three random-effects structures on twelve of those cells at 1,000 replicates.

2.7 Performance measures and computing environment

For each cell we report Power, Type I error, Bias, Empirical SE (calibrated scale), Coverage of the nominal 95% Wald CI, MSE, non-convergence rate, and the MCSE of each, following Morris et al.[1, Tables 5-6]. We chose $n_{sim} = 500$ so that the MCSE on power at $p = 0.5$ would not exceed 0.022; at $n_{sim} = 500$ the two-MCSE band around a nominal 0.05 rejection rate is approximately $[0.031, 0.069]$. No simulation runs while this manuscript renders. All specifications are fit to the same simulated dataset within each cell (common random numbers), so every pairwise contrast carries reduced Monte Carlo variance relative to independent draws. Exact file paths and versions are given in the Reproducibility section.

2.8 Corrections to the simulation code during the study

Two defects in the simulation driver were found and corrected while this study was in progress, and a third change in policy followed from Stage 5. Each is stated here so that the reader can tell which results it affects.

1. **Pre-exposure coding of the exposure-weighted predictor (corrected 2026-10-06).** Before any exposure, $D_{bc,it}$ was coded as $\phi(0) = 1$ rather than 0, so a patient on the placebo-first path of the crossover design was treated as fully exposed at every visit. The same correction replaced G4's REML AIC comparison with the ML comparison and REML refit of Section 2.4. A paired check fitting each simulated trial under the old and new coding (`26-co-dbc-fix-paired.R`) showed that the entire collapse of the exposure-weighted family under the crossover design reported in earlier drafts was this defect: with the correction, G3 and G4 match G1 under CO. Hybrid cells, where no patient starts on placebo, were unaffected.
2. **Cross-factor correlation constant (corrected 2026-10-21).** The documented data-generating process uses a single cross-component correlation $c_1 = 0.2$ scaled by $\rho^{|w_t - w_s|}$ at every lag; the driver had used 0.1 at lags greater than zero. The Stage 1 factorial was rerun in full under the corrected constant, and every Stage 2 to 5 comparison was run after it. The one result retained from before this correction is the 3,000-replicate size diagnostic of Section 3.3, a Hybrid null cell whose conclusion the corrected-code null cells reproduce.
3. **Non-positive-definite covariance matrices are refused, not repaired (policy change, 2026-10-26).** The published code repairs a non-positive-definite covariance matrix with `corpcor::make.positive.definite` and continues; the refined process recommends stopping. Stage 5 found that a prognostic confound at strength 0.6 produced such a matrix and that the repair had removed part of the interaction signal, so the driver now refuses such cells by default. Section 3.5 gives the evidence and the closed-form feasibility bound that replaces trial and error.

Table 2: Stage 1: performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is mean(estimate $-\ \theta_{\text{true}}$) on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$; see the estimand-mismatch caveat in the text for G1/G2. Empirical SE is the standard deviation of the point estimates across replicates. Coverage is the proportion of nominal 95% Wald CIs containing θ_{true}. MSE is bias^2 + Empirical SE^2. Non-conv is the fraction of replicates where 'nlme::lme()' failed to converge. MCSEs follow Morris et al. (2019, Table 6).

t1/2	Specification	Power (MCSE)	Bias (MCSE)	Empirical SE	Coverage (MCSE)	MSE (MCSE)	Non-conv
0.0	Unadjusted	0.994 (0.003)	-0.021 (0.031)	0.699	0.990 (0.004)	0.489 (0.029)	0.000
0.0	Lag-adjusted	0.994 (0.003)	-0.022 (0.031)	0.698	0.990 (0.004)	0.487 (0.029)	0.000
0.0	Exposure-weighted	0.994 (0.003)	-0.021 (0.031)	0.699	0.990 (0.004)	0.489 (0.029)	0.000
0.5	Unadjusted	0.946 (0.010)	+0.618 (0.034)	0.760	0.938 (0.011)	0.959 (0.055)	0.000
0.5	Lag-adjusted	0.944 (0.010)	+0.619 (0.034)	0.762	0.936 (0.011)	0.964 (0.055)	0.000
0.5	Exposure-weighted	0.962 (0.009)	+0.069 (0.039)	0.873	0.970 (0.008)	0.766 (0.052)	0.000
1.0	Unadjusted	0.806 (0.018)	+1.143 (0.036)	0.799	0.790 (0.018)	1.943 (0.092)	0.000
1.0	Lag-adjusted	0.808 (0.018)	+1.143 (0.036)	0.808	0.786 (0.018)	1.959 (0.092)	0.000
1.0	Exposure-weighted	0.888 (0.014)	-0.084 (0.047)	1.062	0.964 (0.008)	1.135 (0.074)	0.000

The exploratory sensitivity blocks of earlier drafts (within-factor autocorrelation, analyst-truth half-life mismatch, dropout, effect-size curve, and single-cell comparisons of G4, G5 and the CR2 variants) all predate both corrections and are no longer part of this manuscript. Their material remains in the supplement and the archived drafts.

3 Results

Results are reported in the order the model was built. Each stage states the question it answers, the data it uses, and what it passes to the next stage. The Monte Carlo standard error on power does not exceed 0.022 at $p = 0.5$, and differences smaller than this should not be over-interpreted.

3.1 Stage 1: representing residual exposure

Stage 1 asks which of the three representations of residual exposure recovers the power that carryover removes. Table 2 reports performance at the reference cell (exponential DGP, Hybrid, $N = 70$, $c_{bm} = 0.45$), where non-convergence was zero. Bias, MSE and coverage are computed against Exposure-weighted's calibrated target, $\theta_{\text{true}} = -c_{bm}\sigma_{BR} = -3.6$, while Unadjusted and Lag-adjusted estimate a different coefficient, bm:D_{it}; their apparent bias and under-coverage reflect that estimand mismatch, not a defect in the estimators, and those columns are comparable within the G1/G2 pair but not across to G3. For the three-way comparison we rely on power and Type I error, properties of the test rather than of the coefficient tested.

Three findings carry the stage, and Figure 1 shows all three.

The lagged-treatment term does nothing. The Unadjusted and Lag-adjusted curves lie on top of one another in every panel and at every half-life. Across all 216 cells of the

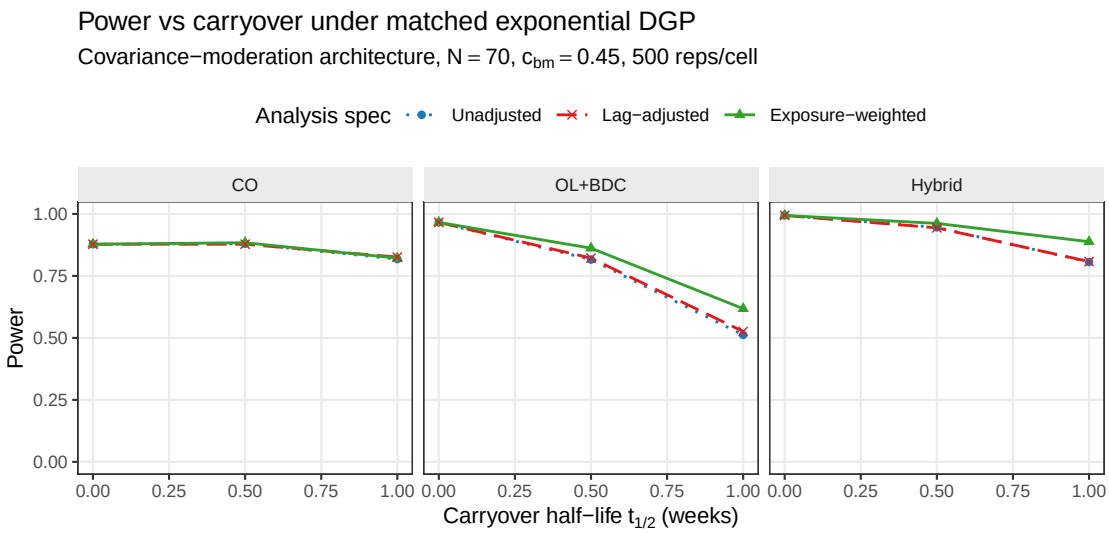


Figure 1: Stage 1: power versus carryover half-life under the exponential DGP, by analysis specification and trial design ($c_{bm} = 0.45$, $n_{sim} = 500$).

covariance-moderation slice the mean power difference is +0.002 (largest absolute difference 0.022, against an MCSE of about 0.02), and mean bias, MSE and coverage agree to three decimal places. Since the two share an estimand, are fitted to the same simulated datasets, and differ by exactly one nuisance term, this is as clean a null as the design can produce. The mechanism follows from where the term sits: it enters only as a main effect, so the tested interaction remains bm:D_{it} and the attenuation is left untouched, and it marks a single occasion per path, so it cannot describe a graded decline across a washout. G2 is carried no further.

Exposure weighting recovers most of the loss. At the reference cell power falls from 0.994 with no carryover to 0.962 at $t_{1/2} = 0.5$ and 0.888 at $t_{1/2} = 1$ week under G3, against 0.806 for G1 at $t_{1/2} = 1$: an advantage of 8 points, about four MCSEs. The advantage is zero at $t_{1/2} = 0$, where there is no carryover to model, and grows with the half-life.

The advantage is design-dependent but never negative. At $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1$, G3 leads G1 by 11 points under OL+BDC (0.618 against 0.512), the design with the fewest post-discontinuation occasions and so the most to gain from weighting them, and by the 8 points above under Hybrid. Under the crossover design the two are indistinguishable (0.822 against 0.818): CO's first off-drug occasion falls 2.5 weeks after discontinuation, where less than a fifth of the effect persists, so there is little residual exposure to represent. Earlier drafts of this manuscript reported G3 as markedly inferior to G1 under CO; that result was the pre-exposure coding defect of Section 2.8 and does not survive its correction.

Under the null the model-based test is conservative for all three: the mean rejection rate across the exponential null cells is 0.038 (range 0.016 to 0.058) against nominal 0.05, and it is conservative to the same degree for each specification, so it does not disturb the comparison among them. Stage 3 takes this up. Stage 1 passes forward G3, with one caveat: its half-life was set equal to the data-generating one, an oracle no analyst has.

Table 3: Stage 2: power of the binary indicator (G1), the exposure-weighted predictor at the true half-life (G3) and the exposure-weighted predictor at the AIC-selected half-life (G4), model-based inference, on the 30-cell corrected grid (exponential DGP, $n_{sim} = 500$, same simulated trial for all three each replicate). Two-MCSE half-width is about 0.04 at $p = 0.5$.

Design	N	c_{bm}	$t_{1/2}$	G1	G3	G4
CO	35	0.45	0.0	0.582	0.582	0.582
CO	35	0.45	0.5	0.576	0.572	0.580
CO	35	0.30	1.0	0.280	0.280	0.294
CO	35	0.45	1.0	0.514	0.524	0.532
CO	70	0.45	0.0	0.880	0.880	0.878
CO	70	0.45	0.5	0.874	0.874	0.874
CO	70	0.30	1.0	0.508	0.518	0.522
CO	70	0.45	1.0	0.832	0.852	0.844
Hybrid	35	0.45	0.0	0.826	0.826	0.816
Hybrid	35	0.45	0.5	0.656	0.678	0.688
Hybrid	35	0.30	1.0	0.212	0.250	0.276
Hybrid	35	0.45	1.0	0.450	0.524	0.534
Hybrid	70	0.45	0.0	0.992	0.992	0.990
Hybrid	70	0.45	0.5	0.960	0.960	0.962
Hybrid	70	0.30	1.0	0.448	0.518	0.534
Hybrid	70	0.45	1.0	0.790	0.886	0.876
OL+BDC	35	0.45	0.0	0.694	0.694	0.700
OL+BDC	35	0.45	0.5	0.474	0.506	0.523
OL+BDC	35	0.30	1.0	0.134	0.164	0.182
OL+BDC	35	0.45	1.0	0.248	0.330	0.344
OL+BDC	70	0.45	0.0	0.958	0.958	0.960
OL+BDC	70	0.45	0.5	0.820	0.842	0.842
OL+BDC	70	0.30	1.0	0.276	0.338	0.346
OL+BDC	70	0.45	1.0	0.544	0.638	0.639

3.2 Stage 2: removing the analyst's half-life

Stage 2 asks whether choosing the half-life from the data by AIC (G4) recovers the power an oracle half-life gives G3. Table 3 reports both, with G1 as the benchmark, on the corrected 30-cell grid of Section 2.6, all fit to the same trial each replicate.

G4 matches G3 within Monte Carlo error at every cell, including the cells where G3 holds the largest advantage over G1 (Hybrid and OL+BDC at $t_{1/2} = 1$), and it does so without any pharmacokinetic input from the analyst. Under CO, where the three differ by at most a few points, G4 is again indistinguishable from G3. The earlier finding that G4's selection drifted to the largest candidate half-life under CO and collapsed its power was a consequence of the same coding defect that affected G3 there, and of comparing REML fits across candidates; both are corrected (Section 2.8). Stage 2 therefore passes G4 forward as the representation of residual exposure, with the question of whether its selection step costs anything in size left to Stage 3.

Table 4: Stage 3: conservatism of the model-based interaction test and its cluster-robust correction at the null cell ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, 3,000 replicates). κ is the calibration ratio of Section 2.4; $\kappa > 1$ is a conservative test. Run before the cross-factor constant correction of Section 2.8.

Specification	κ model	Type I model	κ CR2	Type I CR2
Unadjusted	1.06	0.036	0.98	0.048
Lag-adjusted	1.05	0.039	0.98	0.049
Exposure-weighted	1.10	0.029	1.00	0.045

Table 5: Stage 3: Type I error at the eight null cells of the corrected grid ($c_{bm} = 0$, $t_{1/2} = 1$ week, exponential DGP, $n_{sim} = 500$). Each model-based specification is followed by its CR2 variant (G1/G6, G3/G8, G4/G9). E0 is the published analysis. Two-MCSE band around 0.05: $[0.031, 0.069]$.

Design	N	G1	G6	G3	G8	G4	G9	E0
CO	35	0.030	0.046	0.034	0.052	0.042	0.058	0.150
CO	70	0.046	0.054	0.046	0.050	0.050	0.056	0.128
Hybrid	35	0.042	0.066	0.030	0.044	0.042	0.060	0.070
Hybrid	70	0.032	0.046	0.034	0.054	0.038	0.058	0.060
OL+BDC	35	0.048	0.074	0.048	0.052	0.044	0.056	0.144
OL+BDC	70	0.030	0.046	0.030	0.048	0.032	0.050	0.106

3.3 Stage 3: a correctly sized test

Stage 3 asks whether the interaction test is correctly sized and, if not, how to make it so without disturbing the ranking. Two findings from Stage 1 motivate it: the model-based test rejected at about 0.037 under the null, and G4 adds a selection step that could in principle inflate size.

Table 4 diagnoses the conservatism at a Hybrid null cell with 3,000 replicates, enough to resolve the calibration ratio κ of Section 2.4. All three Stage 1 specifications are conservative under the model-based test, most so G3 ($\kappa = 1.10$, Type I 0.029), and the CR2 standard error restores κ to near one and Type I error to nominal for all three. This diagnostic was run before the cross-factor constant correction (Section 2.8); the corrected-code null cells below reproduce its conclusion.

Table 5 then reports Type I error on the corrected grid for each specification with and without CR2, at all eight null cells, with E0 alongside. Three things are visible. The model-based G1, G3 and G4 sit at 0.03 to 0.05 throughout; their CR2 variants sit at 0.04 to 0.07, inside the two-MCSE band at every cell except G6 at OL+BDC, $N = 35$ (0.074, at its edge). G9 is within 0.016 of G8 at every cell and never above 0.060, so the half-life selection step costs nothing detectable in size once CR2 is applied; a 2,000-replicate check at the Hybrid null (run before the constant correction) gave 0.038 for G4 and 0.052 for G9. And E0, the published analysis, rejects at 0.06 to 0.15 at every null cell, far outside the band under CO and OL+BDC and at its edge under Hybrid.

With size settled, Figures 2 and 3 show power for all nine G-family specifications on the

corrected grid, in the style of Hendrickson et al.'s Figure 4, across the biomarker-effect axis (Panel A) and the carryover axis (Panel B).

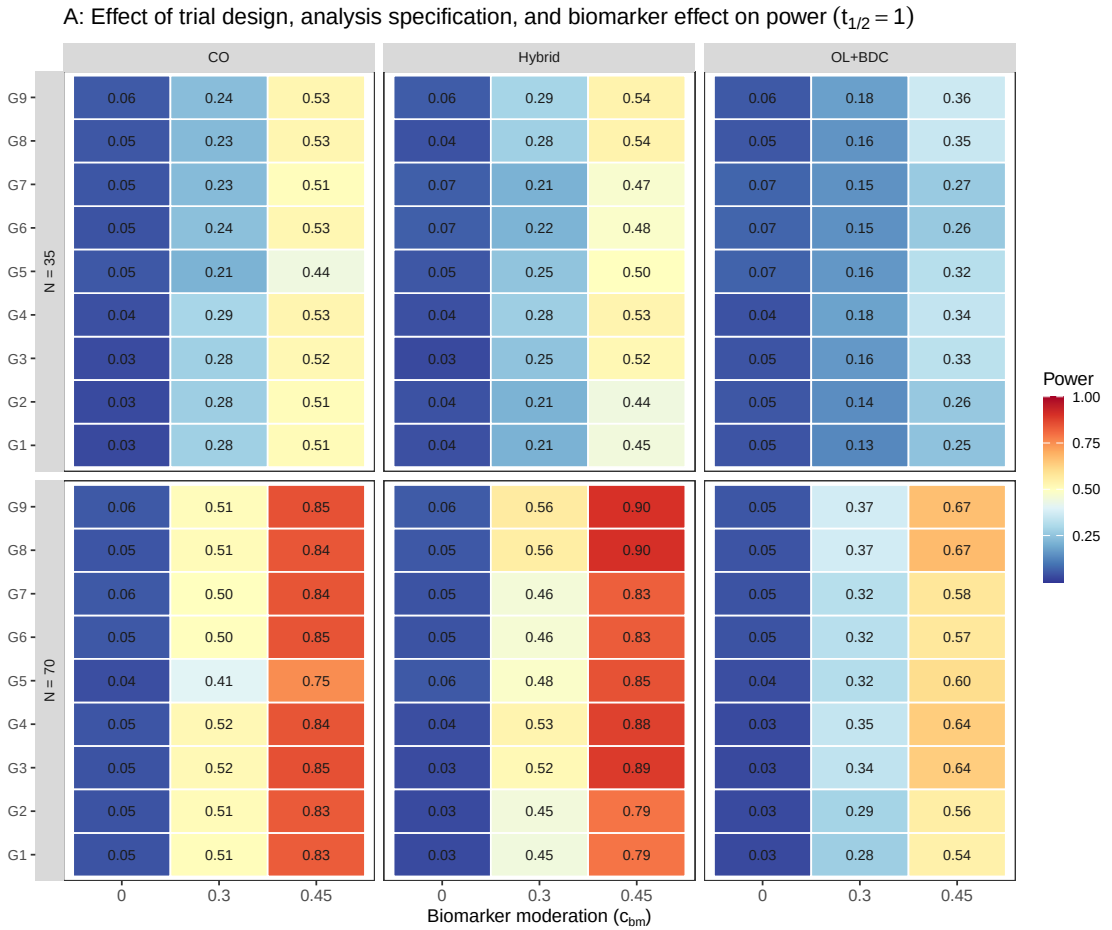


Figure 2: Stage 3: power by trial design (columns), N (row blocks) and analysis specification (rows within block) across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, corrected simulation code ($n_{sim} = 500$). The $c_{bm} = 0$ column is Type I error.

Two features matter for what follows. First, CR2 recovers power as well as size: at the Hybrid reference cell G9 reaches 0.900 against 0.876 for G4, and G8 0.904 against 0.886 for G3. Second, G9 and G8 lead in a near-tie at every Hybrid and OL+BDC alternative cell, at both sample sizes and all three half-lives, while G6 and G7 track G1 and G2 below them; under CO all nine are within a few points of one another except G5. G9 therefore enters Stage 4 as the leading specification: the same power as the oracle-half-life G8, nominal size, and no half-life required of the analyst.

3.4 Stage 4: external comparators on the same trials

Stage 4 asks whether the originally published analysis (E0) or the richer expectancy-term family (F1, F1A0B) does better than G9 on the same simulated trials, across the scenario

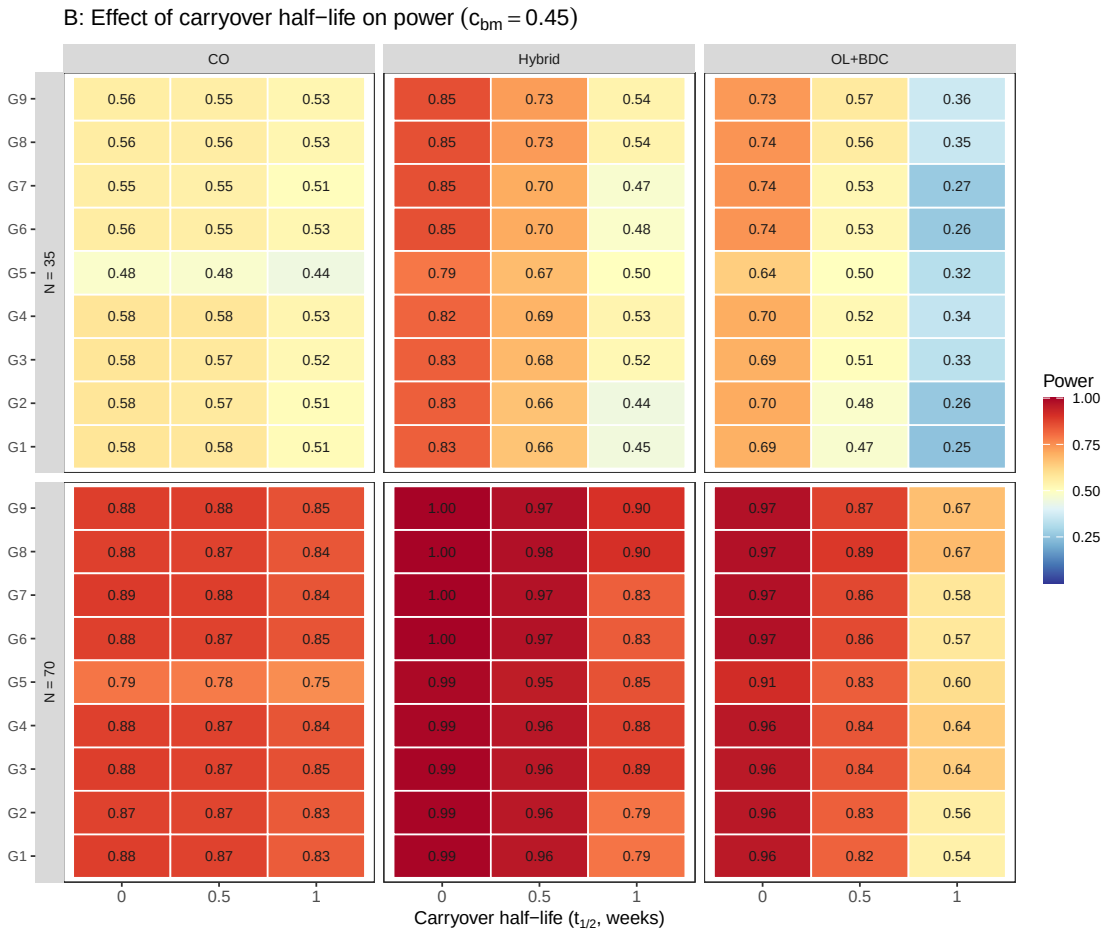


Figure 3: Stage 3: power across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, corrected simulation code ($n_{sim} = 500$).

Table 6: Stage 4: Type I error ($c_{bm} = 0$) at nine null scenarios on Hybrid ($N = 70$, $t_{1/2} = 0.5$ week unless stated; OL+BDC row is that design), $n_{sim} = 500$, same trial for every specification each replicate. The prognostic rows set one pathway parameter to 0.3. Two-MCSE band around 0.05: $[0.031, 0.069]$.

Cell	G1	G3	G4	G8	G9	E0	F1	F1A0B
Hybrid	0.034	0.032	0.034	0.046	0.048	0.060	0.058	0.046
OL+BDC	0.030	0.036	0.036	0.048	0.060	0.100	0.050	0.038
prog-BL 0.3	0.034	0.032	0.034	0.046	0.048	0.062	0.059	0.047
prog-TV-const 0.3	0.034	0.034	0.036	0.046	0.048	0.066	0.061	0.048
prog-TV-grow 0.3	0.034	0.032	0.034	0.046	0.048	0.060	0.058	0.046
prog-PB 0.3	0.034	0.030	0.032	0.048	0.052	0.074	0.058	0.046
$t_{1/2} = 2$ wk	0.034	0.038	0.042	0.054	0.062	0.066	0.059	0.047
Weibull $k = 2$ data	0.032	0.032	0.032	0.046	0.052	0.060	0.058	0.058
mean moderation	0.034	0.032	0.034	0.046	0.048	0.060	0.058	0.046

Table 7: Stage 4: power at six alternative scenarios (same specifications and n_{sim} as the null table).

Cell	G1	G3	G4	G8	G9	E0	F1	F1A0B
Hybrid, $c_{bm} = 0.45$	0.934	0.950	0.950	0.972	0.968	0.932	0.952	0.950
Hybrid, $c_{bm} = 0.25$	0.510	0.542	0.544	0.574	0.584	0.546	0.538	0.500
OL+BDC, $c_{bm} = 0.45$	0.842	0.842	0.825	0.871	0.863	0.866	0.848	0.812
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	0.180	0.242	0.265	0.257	0.281	0.284	0.192	0.228
Weibull $k = 2$ data, $c_{bm} = 0.25$	0.630	0.610	0.631	0.655	0.661	0.612	0.652	0.614
mean moderation, $c_{bm} = 0.25$	0.642	0.642	0.655	0.658	0.669	0.622	0.658	0.633

cells of Section 2.6. Table 6 gives Type I error at the nine null scenarios and Table 7 power at the six alternatives, with G1, G3, G4 and G8 retained for reference.

E0 is liberal everywhere. The published analysis rejects at 0.060 to 0.100 across the nine null scenarios, never at or below nominal, reaching 0.100 under OL+BDC. Read with Table 5, where it reaches 0.15 under CO, the most direct interpretation is that dropping the residual correlation structure, the one feature every G-family specification shares and that postdates the published code, is itself a source of Type I error inflation under this data-generating process, worst where within-patient spacing is least regular. Its power at the alternatives is correspondingly not comparable.

G9 holds size without an expectancy term. All four prognostic pathways leave G9 at 0.048 to 0.052, inside the band, as they do G8 and the F-family. The expectancy terms were introduced to protect against exactly these pathways; at this strength the G-family does not need them. Stage 5 doubles the strength.

G9 loses no power to the F-family and gains where the half-life is wrong. At every alternative G9 is at or above both F1 and F1A0B, the bootstrap-calibrated member. The largest difference is the half-life misspecification cell, true $t_{1/2} = 2$ weeks against F1's fixed 0.2-week policy value: G9 gives 0.281 against 0.192 for F1 and 0.228 for F1A0B. This is the scenario a data-chosen half-life exists for, and G9's selection over $\{0.25, 0.5, 1, 2\}$ reaches the true value where F1's fixed value cannot. Under a Weibull $k = 2$ data-generating decay

Table 8: Stage 5: Type I error of G9 and F1A0B under each prognostic pathway at confound strength 0.3 and 0.6, with no true interaction ($c_{bm} = 0$), Hybrid, $t_{1/2} = 0.5$ week, $n_{sim} = 500$. Column groups, left to right: G9 at $N = 35$, G9 at $N = 70$, F1A0B at $N = 35$, F1A0B at $N = 70$. The strength-0.6 cells on prog-BL, prog-PB and prog-TV-const were generated under a repaired covariance matrix (see text).

	G9, $N = 35$		G9, $N = 70$		F1A0B, $N = 35$		F1A0B, $N = 70$	
Pathway	0.3	0.6	0.3	0.6	0.3	0.6	0.3	0.6
prog-BL	0.048	0.044	0.044	0.034	0.048	0.048	0.047	0.035
prog-TV-const	0.050	0.048	0.046	0.034	0.042	0.046	0.044	0.030
prog-TV-grow	0.046	0.046	0.046	0.034	0.046	0.046	0.046	0.034
prog-PB	0.046	0.044	0.046	0.034	0.050	0.048	0.044	0.034

Table 9: Stage 5: power of G9 and F1A0B with a true interaction ($c_{bm} = 0.45$) and a co-occurring prognostic confound, same layout as the size table. The no-confound reference for G9 is 0.759 at $N = 35$ and 0.969 at $N = 70$ (1,000 replicates). Strength-0.6 cells on prog-BL, prog-PB and prog-TV-const are a data-generating artifact, not a confound effect (see text).

	G9, $N = 35$		G9, $N = 70$		F1A0B, $N = 35$		F1A0B, $N = 70$	
Pathway	0.3	0.6	0.3	0.6	0.3	0.6	0.3	0.6
prog-BL	0.782	0.758	0.976	0.960	0.738	0.702	0.950	0.920
prog-TV-const	0.782	0.730	0.968	0.950	0.502	0.482	0.772	0.702
prog-TV-grow	0.778	0.756	0.970	0.966	0.778	0.756	0.970	0.966
prog-PB	0.798	0.741	0.968	0.940	0.597	0.562	0.853	0.794

analyzed as exponential, and under the mean-moderation architecture, G9 is likewise the most powerful or tied most powerful specification (0.661 and 0.669), so the Stage 1 advantage of exposure weighting is not specific to the exponential, covariance-moderation case at this effect size. F1A0B's bootstrap calibration costs a few points of power relative to F1 at every alternative, the price of guarding against a selection-induced inflation that, for G9 with CR2, does not occur.

Stage 4 therefore passes G9 forward unchanged: nothing outside the family is better sized or more powerful on the same trials.

3.5 Stage 5: prognostic confounds, feasibility, and random effects

Stage 5 asks whether G9 survives a prognostic biomarker at twice the strength of Stage 4, with and without a true interaction present, and whether a richer random-effects structure adds anything. It also reports a finding about the simulation itself.

3.5.1 Prognostic confounds at two strengths

Table 8 gives G9's Type I error, and Table 9 its power at $c_{bm} = 0.45$, across the four pathways at strengths 0.3 and 0.6 and $N \in \{35, 70\}$ on Hybrid, with F1A0B alongside as the expectancy-term comparator.

Table 10: Stage 5: positive definiteness of the raw joint covariance matrix on the four Hybrid paths, and the population on-drug minus off-drug slope contrast of the outcome on the biomarker before and after the silent repair, for the prognostic cells at $c_{bm} = 0.45$ and the strength-0.6 null cells (computed from the matrix, no simulation). The contrast is the population value of the interaction term.

Pathway	Strength	c_{bm}	Paths repaired	Min. eigenvalue	Contrast, raw	Contrast, repaired
none	0.0	0.45	0	0.180	-3.26	-3.26
prog-BL	0.3	0.45	0	0.145	-3.26	-3.26
prog-BL	0.6	0.00	0	0.184	0.00	0.00
prog-BL	0.6	0.45	4	-0.110	-3.26	-3.00
prog-PB	0.3	0.45	0	0.010	-3.40	-3.40
prog-PB	0.6	0.00	4	-0.240	-0.27	-0.23
prog-PB	0.6	0.45	4	-0.642	-3.53	-2.38
prog-TV-const	0.3	0.45	0	0.011	-3.26	-3.26
prog-TV-const	0.6	0.00	4	-0.354	0.00	0.00
prog-TV-const	0.6	0.45	4	-0.853	-3.26	-1.81

G9's Type I error is inside the band at every one of the sixteen null cells, including strength 0.6, and its power at strength 0.3 is at or above the no-confound reference on every pathway. prog-TV-grow, a mean shift absorbed exactly by the biomarker main effect, is flat at both strengths as expected. At strength 0.6 on prog-PB and prog-TV-const, however, G9 lost 17 to 26 points of power, and so did F1A0B and every other specification fitted to the same trials. The uniformity of the loss across fourteen specifications of three families was the first sign that it lay upstream of the analysis.

3.5.2 The strength-0.6 loss is a data-generating artifact

A direct check of the covariance matrix (Table 10) shows why. Each strength-0.6 cell on prog-PB and prog-TV-const, and prog-BL at $c_{bm} = 0.45$, has a joint correlation matrix that is not positive definite on any of Hybrid's four paths. Because the driver then ran with silent repair, `corpcor::make.positive.definite` replaced the matrix by a nearby positive-definite one before any data were drawn. The repair shrinks the population on-drug minus off-drug slope of the outcome on the biomarker, the quantity the interaction term estimates, from about -3.3 to -2.4 (prog-PB) and -1.8 (prog-TV-const), and the fitted G9 estimates at those cells match the repaired values (-2.5 and -2.0 at $N = 35$) while their empirical standard errors fall rather than rise. The power loss is the repair removing a third to a half of the interaction from the generated data; there was nothing for any analysis model to recover. Every strength-0.3 cell is positive definite, which is why none shows a loss.

3.5.3 The feasibility ceiling for a prognostic coupling

The condition that failed is computable in closed form. Write R_0 for the joint correlation matrix with the biomarker removed, which depends on no biomarker coupling, and r for the biomarker row, which is a sum of fixed patterns scaled by the couplings, $r = c_{bm}\, g + c_{pb}\, p +$

Table 11: Stage 5: exact feasibility ceilings for each coupling, alone and given a fixed predictive coupling, Hybrid design (minimum over paths), $t_{1/2} = 0.5$ week, $\rho = 0.7$, $c_1 = 0.2$. Each ceiling is verified by the smallest eigenvalue of the full correlation matrix being positive at 0.999 times and negative at 1.001 times the stated value.

Coupling	Given	Ceiling
c_{bm}	alone	0.518
c_{pb}	alone	0.513
c_{tv}	alone	0.513
c_{bl}	alone	1.000
c_{pb}	$c_{bm} = 0.3$	0.451
c_{tv}	$c_{bm} = 0.3$	0.451
c_{bl}	$c_{bm} = 0.3$	0.815
c_{pb}	$c_{bm} = 0.45$	0.306
c_{tv}	$c_{bm} = 0.45$	0.306
c_{bl}	$c_{bm} = 0.45$	0.496

$c_{tv}\, q + c_{bl}\, b$. By the Schur complement on the biomarker row, the joint matrix is positive definite if and only if R_0 is and

$$r^\top R_0^{-1} r = c^\top A\, c < 1, \qquad A_{jk} = x_j^\top R_0^{-1} x_k, \quad x \in \{g, p, q, b\},$$

the biomarker's squared multiple correlation on everything else. The feasible region is the interior of an ellipsoid, and for fixed values of the other couplings the largest feasible value of any one is the positive root of a quadratic. The design's ceiling is the minimum over its paths. Table 11 gives the ceilings for the configuration used here, each verified by the smallest eigenvalue of the full matrix just below and just above it.

At $c_{bm} = 0.45$ the largest feasible prog-PB or prog-TV-const coupling is 0.306, so strength 0.3 sat just inside the boundary (which is the smallest eigenvalue of 0.010 in Table 10) and strength 0.6 was infeasible at every c_{bm}. Three consequences follow for the study and for anyone designing a similar one. The strength-0.6 power numbers must not be read as the cost of a strong confound; the valid finding is that G9 loses no power to a prognostic confound at any feasible strength tested. The driver now refuses a non-positive-definite matrix by default (Section 2.8), with one carve-out that is not a repair: a timepoint with expectancy zero has exactly zero placebo-response variance, and the positive-variance sub-block is factored instead. And stress scenarios should be chosen from the ceiling table before a run rather than discovered by trial and error afterwards.

3.5.4 Random-effects augmentation adds nothing

Before the artifact was identified, the strength-0.6 loss had been read as added patient-level variance that a random intercept could not absorb, and G9 was refitted with a random slope on time, on expectancy, and on both, on twelve of the Stage 5 cells at 1,000 replicates. Table 12 shows the result: all three variants match plain G9 to within one point of power and of standard error at every cell, with paired per-replicate differences of at most 0.008, at the cost

Table 12: Stage 5: rejection rate of G9 and its three random-effects variants on twelve cells (Hybrid, $t_{1/2} = 0.5$ week, $n_{sim} = 1{,}000$; $c_{bm} = 0$ rows are Type I error). The strength-0.6 rows were generated under the repaired matrix described above.

Cell	G9	+ slope on t	+ slope on D_e	+ both
none, $c_{bm} = 0.00$, $N = 35$	0.050	0.050	0.050	0.046
none, $c_{bm} = 0.00$, $N = 70$	0.043	0.044	0.043	0.041
none, $c_{bm} = 0.45$, $N = 35$	0.759	0.762	0.766	0.769
none, $c_{bm} = 0.45$, $N = 70$	0.969	0.970	0.969	0.970
prog-PB 0.6, $c_{bm} = 0.00$, $N = 35$	0.060	0.060	0.059	0.060
prog-PB 0.6, $c_{bm} = 0.00$, $N = 70$	0.047	0.046	0.047	0.047
prog-PB 0.6, $c_{bm} = 0.45$, $N = 35$	0.594	0.593	0.593	0.596
prog-PB 0.6, $c_{bm} = 0.45$, $N = 70$	0.860	0.859	0.858	0.858
prog-TV-const 0.6, $c_{bm} = 0.00$, $N = 35$	0.050	0.053	0.051	0.054
prog-TV-const 0.6, $c_{bm} = 0.00$, $N = 70$	0.044	0.044	0.043	0.040
prog-TV-const 0.6, $c_{bm} = 0.45$, $N = 35$	0.489	0.488	0.489	0.485
prog-TV-const 0.6, $c_{bm} = 0.45$, $N = 70$	0.772	0.772	0.773	0.768

of up to 2.5% non-convergence for the three-component structure. This is the expected result once the mechanism is understood, since there was no signal to recover, and it closes the random-effects question for G9 on this design.

4 Discussion

4.1 What each stage established

The sequence of stages was not planned in advance; each was built to answer the question the previous one raised, and the recommendation rests on the chain rather than on any single comparison.

Stage 1 established that the choice of exposure representation is a question of power, not validity, and that only the representation that changes the regressor carrying the interaction recovers any of it. The lagged-treatment term, the classical remedy, changed nothing measurable across 216 cells, and the reason is structural: it enters as a main effect and leaves the attenuated interaction untouched. A practitioner should not report a lagged term as an adjustment for carryover. Exposure weighting recovered most of the loss under the two designs with a blinded-discontinuation phase and was never worse than the binary indicator; the earlier finding that it failed under a crossover design was a coding defect, and we withdraw it.

Stage 2 established that the oracle half-life Stage 1 granted the analyst is not needed. AIC selection over four candidates matches the oracle within Monte Carlo error at every cell of the corrected grid, including under the crossover design once the selection compares ML fits and the predictor is coded correctly before first exposure.

Stage 3 established that the model-based test is conservative by about a quarter of its nominal size, that the conservatism comes from a working covariance that does not match the data,

and that a CR2 cluster-robust standard error with Satterthwaite degrees of freedom restores nominal size for every specification while recovering two to five points of power. It also established that the post-selection risk of data-chosen half-life does not materialize as Type I error inflation under CR2.

Stage 4 established two things about analyses outside the family. The published analysis, which lacks a residual correlation structure, is liberal at every null scenario examined, by a factor of two to three under the crossover and OL+BDC designs; adding `corCAR1` to it, which every G-family specification does, is consequential and should be stated as such. And the expectancy-term family, built to protect against a prognostic biomarker, buys no size that G9 lacks and costs power where the half-life is misspecified, because its fixed-half-life member cannot adapt and its bootstrap-calibrated member pays for a correction G9 with CR2 does not need.

Stage 5 established that G9 holds size under all four prognostic pathways at both strengths, loses no power to a confound at any feasible strength, and gains nothing from random slopes. Its methodological contribution is the identification of a silently repaired non-positive-definite covariance matrix as the source of an apparent robustness failure, and the closed-form feasibility ceiling that lets a stress grid be designed inside the valid region.

4.2 The recommended analysis for a Hybrid trial

For a Hybrid aggregated N-of-1 trial under covariance moderation we recommend G9:

$$\mathrm{Sx}_{it} \sim bm_i + t + D_{bc,it}(\hat{t}_{1/2}) + bm_i{:}D_{bc,it}(\hat{t}_{1/2}),$$

with a patient random intercept and `corCAR1` residual correlation; $D_{bc,it}$ equal to one on drug, $2^{-t_{sd}/\hat{t}_{1/2}}$ off drug after first exposure and zero before it; $\hat{t}_{1/2}$ chosen by ML-AIC over $\{0.25, 0.5, 1, 2\}$ weeks on the null model and the selected model refitted by REML; and the interaction tested with a CR2 cluster-robust standard error and Satterthwaite degrees of freedom. Three properties recommend it over the alternatives considered. It requires no pharmacokinetic input from the analyst and matches the power of the oracle-half-life predictor. It holds nominal size at every null scenario examined, including after half-life selection and under four prognostic pathways, without an expectancy term. And it is at least as powerful as the richer expectancy-term family at every alternative scenario, with a clear advantage when the true half-life is long.

The recommendation transfers to OL+BDC on the evidence of Stages 1 to 4, where G9 and G8 lead in the same near-tie. Under the crossover design the specifications are within a few points of one another and G9 is not worse than any; the choice there is immaterial on this evidence, and the expectancy-term family is not identifiable at all. Under the mean-moderation architecture G9 led at the one scenario examined, which is encouraging but not a basis for a recommendation; the archived full-scope drafts remain the authority on that architecture.

4.3 What the expectancy terms are for

The G-family omits the design-expectancy covariate that the F-family carries, and Stages 4 and 5 show that G9 does not need it for size under any pathway at the strengths and sample sizes examined. One mechanism deserves a note. The placebo-response amplitude scales with expectancy, which differs between on-drug and off-drug visits in the Hybrid design, so a biomarker correlated with the placebo response acquires a small on/off slope difference by construction: the population contrast under prog-PB is -0.14 at strength 0.3 and -0.27 at 0.6 even with no true interaction, against -3.3 at the alternative. This is precisely what D_e and $bm{:}D_e$ absorb. At $N \leq 70$ it produced no detectable inflation in G9, but it would grow with N and with the coupling, and a trial with a much larger sample or a biomarker known to track placebo responsiveness should check it directly.

4.4 Limitations

The numbers come from a single parameter set calibrated to a prazosin and PTSD trajectory, so absolute power values are illustrative and the ranking, conditional on design, is the more portable result. Dropout was examined only in exploratory blocks that predate the code corrections and only for G1 to G3; there, power collapsed for all three alike as dropout rose, which suggests retaining patients matters more than any choice among specifications, but G9's behavior under dropout has not been measured. The decay-shape and half-life misspecification checks of Stage 4 are single cells at one effect size each. The feasibility ceilings are specific to the Hybrid design at $t_{1/2} = 0.5$ week, $\rho = 0.7$ and $c_1 = 0.2$, and must be recomputed for any other configuration; the closed form makes that immediate. Size at the strength-0.6 prognostic cells was measured under a repaired matrix and is a result about a modified data-generating process. The exploratory sensitivity blocks of earlier drafts, which varied autocorrelation, half-life mismatch and effect size one at a time for G1 to G3, were not rerun under the corrected code and are not relied on here. And the sensitivity checks vary one factor at a time, so they can miss interactions between factors.

4.5 Related approaches not evaluated

Three N-of-1 analysis approaches sit outside the family examined and would each require substantial independent development to enter the comparison. Tang and Landes[9] propose formula-based t-tests that correct a single-subject paired-difference test for an estimated serial correlation; they have no biomarker-moderation term. Schmid and Yang[10] describe a Bayesian hierarchical alternative with partial pooling of individual-level treatment, trend and autocorrelation parameters, noting that carryover is tractable only in aggregated analyses like ours. Daza[11] frames N-of-1 inference as a counterfactual time-varying-treatment problem in which carryover is confounding to be controlled rather than a predictor to be modeled. Senn, Julious and Araujo's[12] finding that a simple paired analysis can match carryover-adjusted mixed models motivated the paired-difference specification G5 in earlier drafts; on the corrected grid it trails the mixed-model family under every design and is not pursued.

5 Conclusions

1. **Represent residual exposure in the regressor, not as a nuisance term.** The classical lagged-treatment term never improved on ignoring carryover, on any measure, in any design; an exposure-weighted predictor recovers most of the power carryover removes under the Hybrid and OL+BDC designs and is never worse than the binary indicator.
2. **Let the data choose the half-life.** AIC selection over four candidates, comparing ML fits and refitting the winner by REML, matches the oracle-half-life predictor at every cell and removes the one input the analyst cannot supply.
3. **Use a CR2 cluster-robust standard error.** The model-based test rejects at about 0.037 under the null; CR2 with Satterthwaite degrees of freedom restores nominal size for every specification and recovers two to five points of power, and it absorbs the post-selection risk of the half-life choice.
4. **Do not use the published analysis as is.** Without a residual correlation structure it is liberal at every null scenario, by two to three times nominal under the crossover and OL+BDC designs.
5. **Expectancy terms are not needed for size, and cost power when the half-life is fixed.** G9 holds nominal size under all four prognostic pathways at both strengths examined and is at least as powerful as the bootstrap-calibrated expectancy-term specification at every alternative.
6. **Check feasibility before stressing the data-generating process.** A prognostic coupling that exceeds the closed-form ceiling yields a non-positive-definite matrix; silently repairing it removes part of the interaction and produces a spurious robustness failure in every analysis model at once. The ceiling is a quadratic root and should be computed first.
7. **Recommendation.** For a Hybrid aggregated N-of-1 trial under covariance moderation, use G9: the exposure-weighted predictor at the AIC-selected half-life with CR2 inference. Open items are its behavior under dropout and a rerun of the earlier exploratory blocks under the corrected code.

6 Declarations

Ethics approval and consent to participate. Not applicable; this is a simulation study with no human participants or patient data.

Consent for publication. Not applicable.

Availability of data and materials. The data and code supporting the findings of this study are available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`. Simulation drivers are at `analysis/scripts/carryover-sensitivity/`. Exact paths are listed in the Reproducibility section below.

Competing interests. The authors declare no competing interests.

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7 Reproducibility

This manuscript was rendered inside the project's zzcollab Docker container, with the image hash in `Dockerfile` and the package set in `renv.lock`. The simulations ran under R 4.6.0 on the project's simulation host. The study uses `nlme` for the continuous-time linear-mixed analysis, `clubSandwich` for CR2 inference, `lme4` and `lmerTest` for E0, and `parallel` for the parameter-grid sweep. No simulation code runs during knitting; knitting loads pre-computed summaries only.

Drivers and data, by stage. All paths are under `analysis/scripts/carryover-sensitivity/` (scripts) and `analysis/data/` (summaries).

- Stage 1: `01-run-factorial.R` (540 cells, 500 replicates, master seed 20260415; rerun 2026-10-25 under the corrected code), `02-summarise-grid.R` producing `02-grid-summary.rds`, and `03-render-figures-extra-slim.R` producing Figure 1.
- Stages 2 and 3: `27-run-tier1-g-vs-f-e0.R` (30 cells, 500 replicates, 2026-10-25) producing `carryover-sensitivity/tier1-g-vs-f-e0/summary.rds`, from which Tables 3 and 5 and Figures 2 and 3 (via `12-hendrickson-heatmap.R`) are drawn. The 3,000-replicate size diagnostic of Table 4 is `diagnose-null-type1-cr2.R` (`output/diag-null-type1-cr2.rds`), run before the cross-factor constant correction; the 2,000-replicate check of G4 and G9 cited in Stage 3 is `16-run-sensitivity-s11.R`.
- Stage 4: `06-g-vs-f-comparison.R` (15 cells, 500 replicates, 2026-10-23) producing `carryover-sensitivity/g-vs-f-comparison/summary.rds`.
- Stage 5: `28-run-g9-prognostic-stress.R` (32 cells, 500 replicates, 2026-10-25) and `29-run-g9-random-effects.R` (12 cells, 1,000 replicates, 2026-10-26) producing `carryover-sensitivity/g9-prognostic-stress/summary.rds` and `carryover-sensitivity/g9-random-effects/summary.rds`; `30-check-prognostic-confound-pd.R` and `31-prognostic-coupling-ceiling.R` producing `pd-check.rds` and `coupling-ceilings.rds` in the former directory, each running in seconds with no simulation.
- Corrections: the paired check of the pre-exposure coding defect is `26-co-dbc-fix-paired.R` (`carryover-sensitivity/26-co-dbc-fix-paired-reps500.rds`); the cross-factor constant correction and the stop-on-non-PD default are in `simulation-core.R` and `implementations/tidyverse/R/functions.R`.

The fitting functions for every specification (`fit_spec_s7`, `cr2_extract`, `select_half_life_aic`, `fit_spec_e0`, `fit_f_family`, `simulate_cell_combined`, `simulate_cell_g9_random`) are in `simulation-core.R`. Specifications are stored under `E`-prefixed codes (`E1`, `E2`, `E3`, `E7`, `E9` and their `cr2` variants) in every data file and displayed as `G1`-`G9` at render time by `spec-labels.R`. Output archived before 2026-07-31 stores the specifications as `A1`, `A2`, `A3`; the migration is documented in `analysis/report/NOTATION.md`. The exploratory sensitivity blocks of earlier drafts and their figures are documented in `supplement.Rmd` and the `archive/` directory of

this manuscript, with the scripts that produced them retained under the same directory as above.

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